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ABSTRACT

Nickel oxide (NiO) is a promising p-type semiconductor extensively investigated for applications in photovoltaics, electrochemical energy storage, and gas sensors. Several deposition techniques, including solution processing, sputtering, chemical vapor deposition, and atomic layer deposition (ALD), are adopted for the synthesis of NiO films. Among these techniques, ALD is recognized for the merit of delivering pinhole-free and conformal films, relevant for processing on textured surfaces. In the present study, we have adopted three ALD NiO processes to deliver films differing in stoichiometry, crystallographic, and optoelectrical properties. To this purpose, the Ni(^{t}Bu-MeAMD)$_2$-based thermal ALD process (“$NiO_{Bu\text{-}MeAMD}$”), the Ni(MeCp)$_2$-based plasma-assisted ALD (PA-ALD) process (“NiO_{MeCp}”), and the newly developed PA-ALD process, based on the novel Alanis^{TM} precursor (“NiO_{Alanis}”), are compared. The NiO_{Alanis}-based process exhibits a linear growth on c-Si/ native oxide with a growth per cycle (GPC) of 0.27 ± 0.02 Å/cycle at 150 °C, similar to the NiO_{MeCp} process, whereas the $NiO_{Bu\text{-}MeAMD}$ process is characterized by a higher GPC of 0.43 ± 0.01 Å/cycle. The NiO_{Alanis} films have the lowest Ni-to-O ratio (0.82 ± 0.04) compared to NiO_{MeCp} (0.96 ± 0.05) and $NiO_{Bu\text{-}MeAMD}$ (0.93 ± 0.05). The excess oxygen in the NiO_{Alanis} films points to a larger relative content of Ni^{3+} acceptor states. The presence of Ni^{3+} states contributes to sub-bandgap absorption, leading to a lower transmittance of the NiO_{Alanis} films (~79%) as compared to the NiO_{MeCp} (~84%) and $NiO_{Bu\text{-}MeAMD}$ films (~90%) in the visible range. In parallel, the resistivity of the (~9 nm) NiO_{Alanis} films ($6 \pm 2\,\Omega$ cm) is lower than NiO_{MeCp} ($66 \pm 33\,\Omega$ cm) and $NiO_{Bu\text{-}MeAMD}$ films [$(1.7 \pm 0.1) \times 10^4\,\Omega$ cm], supporting the observation of the presence of a larger Ni^{3+} content. Transmission electron microscopy images reveal the presence of a relatively higher density of grain boundaries in $NiO_{Bu\text{-}MeAMD}$ films that can affect carrier mobility. Furthermore, the PA-ALD NiO films exhibit a higher density (6.9 ± 0.8 g/cm^3) than the thermal ALD $NiO_{Bu\text{-}MeAMD}$ films (5.1 ± 0.5 g/cm^3) due to the ion flux impinging on the surface during the O_2 plasma half-cycle. Our study demonstrates that the ALD process influences the stoichiometry and microstructure of the NiO film, ultimately impacting its electrical and optical properties. Additionally, we reflect on the selection of ALD processes for specific application areas.

I. INTRODUCTION

Nickel oxide thin films are widely investigated due to their applications, among others, in metal halide perovskite and organic solar cells,[1–3] lithium-ion batteries,[4,5] electrocatalysis,[6,7] thin film transistors,[8] gas sensors,[9] and resistivity switching devices.[10] NiO is a promising material in the realm of wide bandgap semiconductors, exhibiting a bandgap exceeding 3.5 eV.[1] Stoichiometric NiO is an insulator ($>10^{13}\,\Omega$ cm).[11] However, as-deposited NiO exhibits p-type conductivity due to the presence of Ni vacancies or excess oxygen.[12] NiO films are also characterized by optical transmittance[13] and demonstrate an antiferromagnetic nature.[14]

NiO thin films have been deposited by various methods such as solution processing,[15,16] sputtering,[17,18] spray pyrolysis,[19] chemical vapor deposition (CVD),[20,21] pulsed laser deposition,[22] and atomic layer deposition (ALD).[1,2,23] Among these deposition techniques, vapor phase deposition methods have an advantage over solution processing due to their ease of scalability. Additionally, solution processing often requires postannealing treatment, which may not be compatible with temperature-sensitive substrates. ALD distinguishes itself from the other vapor phase deposition techniques because the self-limiting nature of the surface reactions enables the growth of conformal thin films, also on textured and high surface area substrates.[24] Recent reports have shown that the adoption of ALD NiO in combination with organic hole transport layers, such as poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine or self-assembled monolayers, in tandem perovskite/c-Si [and copper indium gallium diselenide (CIGS)] devices leads to uniform coverage on textured c-Si and CIGS bottom cells, thereby suppressing shunt pathways and leading to a high yield in device performance.[25,26] Moreover, uniform coverage of thin films on 3D or porous substrates can be achieved by ALD, making it particularly advantageous for applications such as electrocatalysis.[24]

ALD processes of NiO have been developed using several Ni precursors, such as $NiCp_2$,[27–31] $Ni(acac)_2$,[14] $Ni(MeCp)_2$,[1,32,33] $Ni(EtCp)_2$,[34,35] $Ni(^tBu\text{-}MeAMD)_2$,[25,36] etc., with different oxidants (O_2 plasma, H_2O, O_3, or H_2O_2). Table SI in the supplementary material reports an overview of previously developed ALD NiO processes. Precursors, such as $Ni(acac)_2$,[37,38] $Ni(thd)_2$,[39,40] $Ni(acac)_2(TMEDA)$,[38] and $Ni(^tBu_2DAD)_2$,[41] have low volatility below 150 °C, necessitating a higher deposition temperature, as mentioned in Table SI in the supplementary material, to promote film growth. This increases the thermal budget of the process and hampers the application on thermally sensitive substrates and surfaces.[10] To this end, selected works demonstrated ALD processes extending to lower temperatures and employing $Ni(^tBu\text{-}MeAMD)_2$,[25] $Ni(MeCp)_2$,[1] $Ni(EtCp)_2$,[34,35] and nickel amino-alkoxides[42,43] as reported in Table SI in the supplementary material. At the same time, nickel amino-alkoxide precursors, such as $Ni(dmamp)_2$, decompose at temperatures around 200 °C,[43,44] limiting the temperature window.

In the present study, we report on an ALD process based on a novel Ni precursor—Alanis™, which allows processing at a temperature as low as 100 °C with a temperature window of 100–250 °C. Additionally, we compare the Alanis™-based plasma-assisted ALD (PA-ALD) NiO (hereafter referred to as "NiO_{Alanis}") with two ALD NiO previously developed in our group—namely, $Ni(MeCp)_2$-based plasma-assisted ALD NiO and $Ni(^tBu\text{-}MeAMD)_2$-based thermal ALD NiO (hereafter, referred to as "NiO_{MeCp}" and "$NiO_{Bu\text{-}MeAMD}$," respectively). The study includes the ALD process development of NiO_{Alanis} with saturation curves as well as a comparison with the other NiO processes in terms of processing temperature window and uniformity. Moreover, we propose a comparative study of the NiO film properties for the three processes, with the aim of demonstrating the effect of ALD processing and chemical properties on the microstructure, electrical, and optical properties. Additionally, we reflect on how specific material properties can influence the selection of certain ALD processes to fulfill requirements or optimize performance in certain application areas.

II. EXPERIMENTAL DETAILS

A. Atomic layer deposition of NiO

The plasma-assisted ALD process of NiO_{Alanis} is developed in a home-built ALD reactor, described elsewhere.[45] It is a vacuum system equipped with a rotary and turbomolecular pumping unit that can reach a base pressure of 10^{-6} Torr. The wall temperature of the reactor is kept at 100 °C. The ALD process is developed at a table temperature range of 50–350 °C. Alanis™, from Air Liquide, is used as the precursor, and a 100 W inductively coupled plasma (ICP) oxygen plasma is used as the coreactant. The physical properties of the Alanis™ precursor are reported in Sec. C in the supplementary material. The Alanis™ bubbler is kept at 80 °C, and the precursor is vapor-drawn via a delivery line heated at 100 °C. A constant flow of oxygen from the top of the reactor is maintained during the ALD cycle to prevent NiO deposition on the quartz tube near the ICP plasma source. The chamber pressure in the home-built ALD reactor (for both PA-ALD processes) in the presence of O_2 plasma is between 1.4×10^{-2} and 1.5×10^{-2} mbar. The O_2 flow rate is 134 ± 5 SCCM. For Alanis™ dosing time of 1 s or longer, the precursor is dosed in pulses with the two pulses separated by 1 s to allow for a build-up of precursor vapor pressure inside the bubbler. The vapor-drawn process, after investigation of the saturation curves, is illustrated in Fig. 1. In short, one NiO ALD cycle consists of two pulses of 0.5 s Alanis™ dose, which is separated by 1 s, followed by 1 s of precursor purge, 5 s of O_2 plasma exposure, and 2 s of purging.

The NiO_{MeCp} and $NiO_{Bu\text{-}MeAMD}$ processes, along with their corresponding saturation curves, have been previously reported by Koushik *et al.* and Phung *et al.*, respectively.[1,25] The NiO_{MeCp} process is carried out in the same home-built ALD reactor as adopted for the Alanis™-based NiO process. The $Ni(MeCp)_2$ bubbler is kept at 55 °C, and an argon carrier gas is used for the delivery via a delivery line heated to 75 °C. The deposition is carried out at a temperature of 150 °C, and the wall temperature is kept at 100 °C. The $NiO_{Bu\text{-}MeAMD}$ process is carried out in the commercial FlexAL™ MK1 ALD reactor (Oxford Instruments) because precursor saturation could not be achieved in the home-built ALD reactor, as shown in Fig. S2 in the supplementary material, due to the presence of background water. The reactor design and the uniform wall temperature of the FlexAL™ reactor[46] facilitate efficient removal of residual water dose, as compared to the home-built ALD reactor. The $Ni(^tBu\text{-}MeAMD)_2$ bubbler is

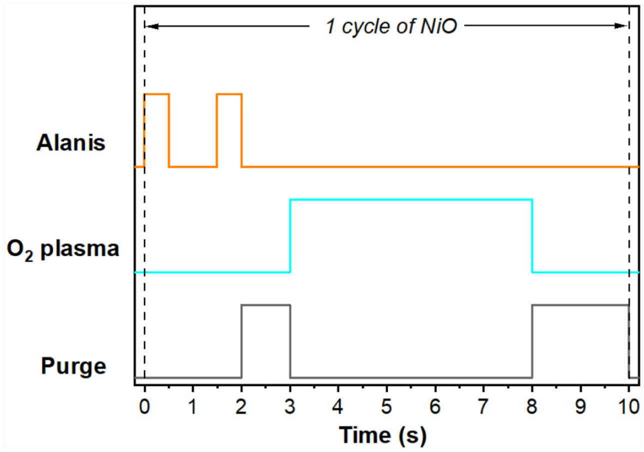

FIG. 1. Schematic of the standard recipe of one cycle of the PA-ALD process of NiO_{Alanis} at 150 °C.

kept at 90 °C, and an argon flow of 100 SCCM through the bubbler is used for delivery at a temperature of 125 °C. The chamber temperature is set at 120 °C, and the deposition is carried out at a table temperature of 150 °C. A short summary of the three NiO ALD processes is shown in Table I and Fig. S1 in the supplementary material.

B. Nio film characterization

The thickness of the NiO films is determined by spectroscopic ellipsometry (SE) (Ellipsometer M2000, J.A. Woollam Co.). The growth of the ALD NiO film is monitored by *in situ* SE (VIS Ellipsometer M2000, J.A. Woollam Co.) on c-Si (100)/~3 nm native oxide substrates. Additionally, the conformality of the NiO film is monitored by SE on an 8″ mapping stage (NIR Ellipsometer M2000, J.A. Woollam Co.). The SE data of the c-Si substrate and the native SiO_2 layer are modeled using the optical constants reported by Herzinger *et al.*[47] The NiO data are fitted by a Cauchy dispersion model in the wavelength range of 400–1000 nm. The thickness nonuniformity is calculated using Eq. (1), where $\sigma_{thickness}$ and $\overline{thickness}$ is the standard deviation and average, respectively, of the thickness of the film measured at several points across the wafer and expressed as a percentage,[48]

$$\text{Non-uniformity} = \left(\sigma_{\text{thickness}} \Big/ \overline{\text{thickness}}\right) \times 100\%. \quad (1)$$

The elemental composition of the NiO films on c-Si/native oxide substrates is analyzed by Rutherford backscattering spectroscopy (RBS) and elastic recoil detection (ERD) using a 3.5 MV Singletron with a 1000 keV He^+ beam at an angle of 170°. Furthermore, the films are characterized by x-ray photoelectron spectroscopy (XPS) measurements performed on the Thermo Scientific KA1066 spectrometer using monochromatic Al K_α x rays with an energy of 1486.6 eV. The data are analyzed by the Avantage software with a Smart background (Shirley background with offset correction) subtraction. The binding energy axis is corrected for sample charging using the adventitious C1s peak at 284.8 eV. The impurity content in the NiO films is evaluated after cleaning the NiO surface by Ar^+ sputtering at 200 eV for 10 s. The Ni-to-O ratio of the films is not reported after the surface cleaning procedure since preferential sputtering of O and reduction to metallic Ni have been observed even at low Ar^+ energies (Fig. S7 in the supplementary material).[49,50]

X-ray diffraction (XRD) is adopted to study the crystal structure of the NiO films on c-Si/native oxide substrates. The measurements are carried out in a Bruker D8 Discover using a Cu k_α ($\lambda = 1.54$ Å) radiation in the range of 30°–80° with a step size of 0.04° and a time per step of 10 s in the grazing incidence (GI) geometry with an incidence angle of 0.4° and using two slits of 0.6 and 0.2 mm for the incident ray. Measurements are also carried out in Bragg–Brentano geometry (θ–2θ scans) in the range of 35°–50° with a step size of 0.4°, a time per step of 10 s, and an omega (ω) angle offset of 3° to suppress the contribution of the Si substrate. X-ray reflectivity (XRR) measurement is also carried out in the same equipment in the range of 0°–6° with a step size of 0.01° and a time per step of 0.1 s. The data are analyzed by the Leptos software using a model consisting of a Si wafer, a thin SiO_2 layer, and a crystalline NiO layer.

Transmission electron microscopy (TEM) measurements are carried out in top-view on NiO films deposited on TEM windows and in cross section on Focused Ion Beam (FIB) prepared lamellas of NiO films deposited on c-Si substrates using a probe-corrected JEOL ARM 200F transmission electron microscope operated at 200 kV and equipped with a 100 mm^2 Centurio silicon drift detector energy dispersive x-ray spectroscopy detector.

The resistivity of the NiO films on the c-Si/450 nm SiO_2 substrate is measured at room temperature using a Signatone four-point probe in combination with a Keithley 2400 source meter. The charge carrier concentration is determined from electrochemical impedance spectroscopy (EIS) by Mott–Schottky analysis. The measurement is carried out in a three-electrode electrochemical cell using a CompactStat (Ivium) potentiostat, and the data are analyzed by the Ivium software. For this purpose, NiO films deposited

TABLE I. Overview of the process parameters of the different ALD processes used in this work.

Process	Precursor	Coreactant	Precursor dosing time (s)	Purge time (s)	Coreactant dose time (s)	Purge time (s)	Temperature (°C)	Reference
NiO_{Alanis}	Alanis^{TM}	O_2 plasma	1	2	5	2	150	This work
NiO_{MeCp}	$Ni(MeCp)_2$	O_2 plasma	4	3	3	1	150	1
$NiO_{Bu\text{-}MeAMD}$	$Ni(^tBu\text{-}MeAMD)_2$	H_2O	4	20	0.05	60	150	25

on glass/ITO substrates are placed into a sample holder with a 1 cm^2 aperture serving as the working electrode. A reversible hydrogen electrode and a counter electrode are used as reference and counter electrodes, respectively. The potential scan of the EIS is carried out in 1.0M Na_2SO_4 solution in the range of 0.2–1.2 V with a step of 10 mV. The EIS curves are obtained in the frequency range of 1 Hz–20 kHz, and the Nyquist plot is modeled by a modified Randles circuit as shown in Fig. S12(a) in the supplementary material. The charge carrier concentration is determined from the Mott–Schottky plots [see Figs. S12(b)–S12(d) in the supplementary material] at 10 kHz applied potential. XPS analysis, carried out to investigate potential changes in the NiO films during electrochemical measurements, does not show major changes as shown in Fig. S13 in the supplementary material.

The transmittance and reflectance of the NiO films are characterized using ultraviolet-visible (UV-Vis) spectroscopy using a PerkinElmer LAMBDA 1050 spectrophotometer with an integrating sphere since it also accounts for scattered and diffused light. For this purpose, the measurements are carried out on NiO deposited on glass slides in the wavelength range of 300–1000 nm with a 2 nm resolution. Ultraviolet photoelectron spectroscopy (UPS) measurements are carried out on NiO films deposited on conductive glass/ITO substrates in a VG EscaLab II system (Thermo Fischer Scientific Inc.) with a base pressure of 10^{-8} Pa using a He I radiation (21.22 eV) and −6 V bias, with a step size of 0.05 eV.

III. RESULTS AND DISCUSSION

A. ALD NiO process with Alanis^{TM} as precursor

The growth of the ALD NiO_{Alanis} film using O_2 plasma, as a coreactant, is investigated. Saturation curves for the precursor and coreactant dosing and purging steps for the Alanis^{TM}-based NiO process, at 150 °C, are determined from *in situ* spectroscopic ellipsometry and reported in Fig. 2. An O_2 plasma of 5 s is used for the precursor saturation curve while varying the Alanis^{TM} dose. A soft saturation is observed for the Alanis^{TM} precursor dosing time [see Fig. 2(a)]. This can be due to slow kinetics at the surface, which can lead to blocked chemisorption sites becoming available as the exposure time increases.[51,52] For the O_2 plasma saturation curve [Fig. 2(c)], the precursor is dosed in pulses twice for a cumulative dosing time of 1 s. It can be seen that there is no growth of the NiO film when the plasma is not ignited, which points to the fact that the Alanis^{TM} precursor does not react with molecular oxygen under the adopted ALD deposition conditions, e.g., low pressure. ALD saturating behavior is observed, within the error, for O_2 plasma half cycles with a growth per cycle (GPC) of 0.27 ± 0.02 Å/cycle. A negligible variation in GPC is observed as a function of the precursor and O_2 plasma purge times, as shown in Figs. 2(b) and 2(d), respectively. A precursor and O_2 plasma purge of 1 and 2 s, respectively, are selected for the process.

The NiO_{Alanis} process has a linear growth on c-Si without nucleation delay, as can be seen in Fig. 3(a), similar to the NiO_{MeCp} process. On the other hand, the $NiO_{Bu\text{-}MeAMD}$ process has a nucleation delay of around 50 cycles on c-Si, followed by linear growth. The delay in growth can be due to the limited number of active sites available for the chemisorption of the precursor molecule onto the substrate surface.[53–56] The GPC of $NiO_{Bu\text{-}MeAMD}$ in the linear growth region is higher than that of NiO_{Alanis} and NiO_{MeCp}, as reported in Table II. The reason behind the higher GPC of the $NiO_{Bu\text{-}MeAMD}$ process is discussed later. The processing time of the $NiO_{Bu\text{-}MeAMD}$ process is longer than the other two processes due to the longer purge time required after the water dosing.

Next, the GPCs of the different ALD NiO processes are investigated at different temperatures, as shown in Fig. 3(b). The GPC of NiO_{Alanis} is relatively stable in the range of 100–250 °C. For temperatures lower than 100 °C, the GPC increases, which can be due to the condensation of the precursor, resulting in a CVD contribution. On the other hand, the increase in GPC around 300 °C might be due to precursor decomposition (see Fig. S3 and Table SII in the supplementary material), thereby limiting the ALD temperature window to a table temperature below 300 °C.[57] The higher GPC of $NiO_{Bu\text{-}MeAMD}$ at low temperatures (50–100 °C) is most likely due to a CVD growth because of the challenge of purging H_2O from the chamber at low temperature. The GPC decreases with an increase in temperature, leading to a GPC of 0.39 ± 0.02 Å/cycle at 200 °C, similar to what has been reported by Hsu *et al.*[36] This can be attributed to the decrease in surface hydroxyl group coverage due to a thermally induced dehydroxylation reaction.[58,59] The process is not carried out at a temperature higher than 250 °C as the Ni $(^{t}Bu\text{-}MeAMD)_2$ precursor has been reported to decompose at 250 °C, resulting in the formation of metallic Ni.[60] The growth of NiO_{MeCp} also decreases with an increase in temperature but remains fairly stable within the temperature window of 150–300 °C. Deposition of these NiO thin films at temperatures of 150 °C or lower enables processing on polyethylene terephthalate substrates, often used in flexible photovoltaics and electronics.[23,61]

The NiO film nonuniformity derived from SE mapping is compared in Fig. S4 in the supplementary material and Table II. The NiO_{Alanis} and NiO_{MeCp} films are deposited on a 4″ c-Si wafer, whereas the $NiO_{Bu\text{-}MeAMD}$ layer is also deposited on an 8″ c-Si wafer because of the possibility of processing large area substrates in the FlexAl^{TM} ALD reactor. The NiO_{Alanis} films exhibit greater nonuniformity compared to NiO_{MeCp} and $NiO_{Bu\text{-}MeAMD}$ films, likely due to the soft saturation behavior of the ALD process. The area near the precursor inlet experiences a higher residence time of the precursor, leading to higher growth. This can cause a thickness gradient along the precursor flow direction, as shown in Fig. S4(a) in the supplementary material.[51,62,63]

B. Film chemical characterization

The chemical composition of the NiO films is reported in Table III and Fig. S5 in the supplementary material. The NiO films are nonstoichiometric with an excess of oxygen content, resulting in a Ni-to-O ratio of less than 1 (reported in Table III). The presence of excess oxygen or Ni vacancies in the lattice leads to the conversion of two Ni^{2+} ions to Ni^{3+} ions to maintain charge neutrality, as shown in Fig. 4.[12,64,65] The p-type conductivity of the NiO film arises due to the formation of the Ni^{3+} acceptor states. Hole transport in NiO occurs when the Ni^{3+} states accept a 3d electron from a neighboring Ni^{2+} state or 2p electron from an adjacent O^{2-} anion, as indicated in Fig. 4 by arrows, creating a hole in the 3d band of Ni^{2+} or 2p band of O^{2-}, respectively.[65,66] XPS measurements are carried out to provide more information about the

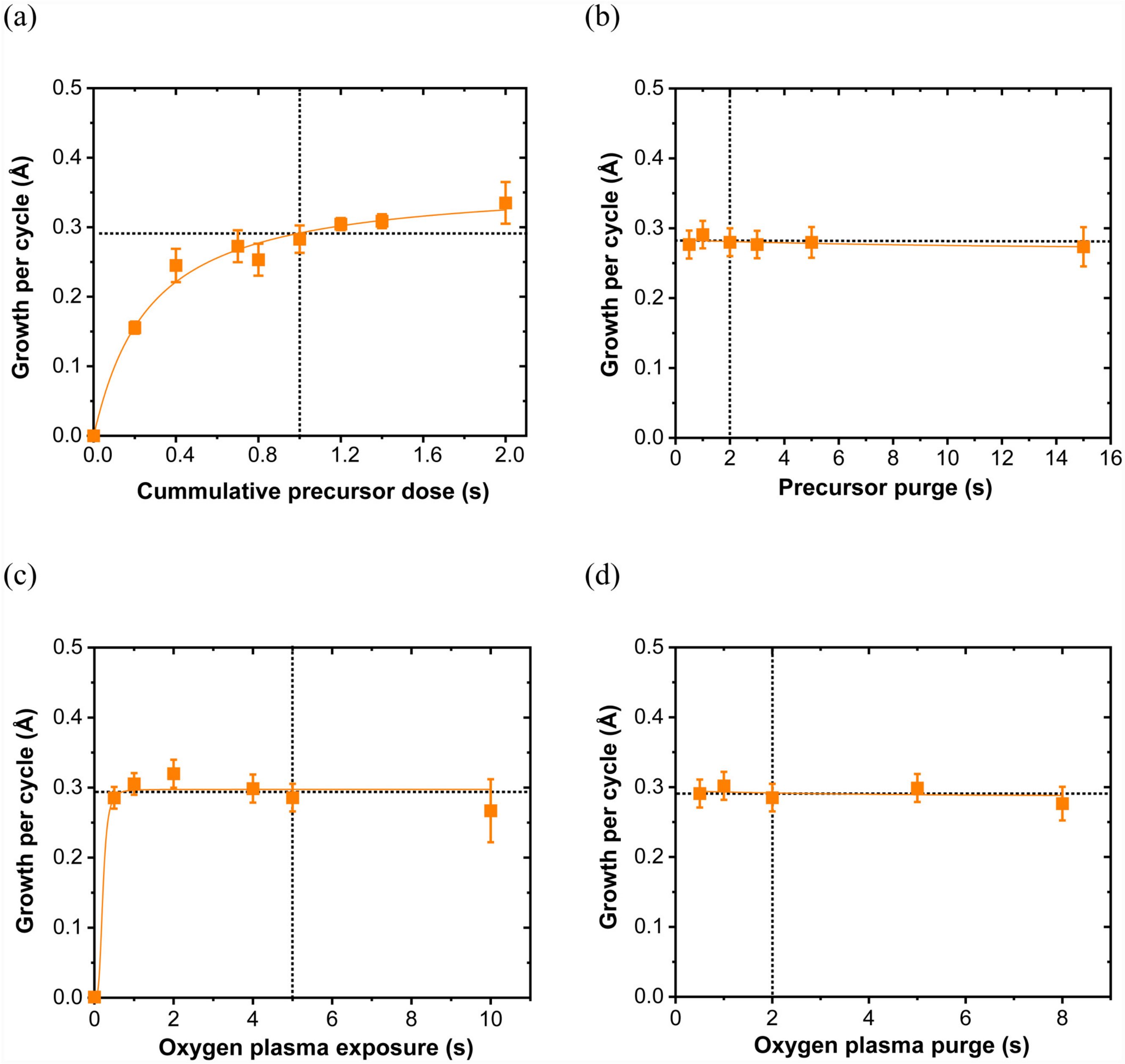

FIG. 2. Saturation curves: GPC as a function of (a) precursor (Alanis™) dosing, (b) precursor (Alanis™) purge, (c) oxygen plasma exposure, and (d) oxygen plasma purge at a deposition temperature of 150 °C. The solid lines through the data points in (a) and (c) are Langmuir isotherm fits. The determined saturation GPC value is indicated by the dotted black line.

chemical state of the elements. Figure S6(a) in the supplementary material shows the normalized Ni2p spectrum split into $2p_{3/2}$ and $2p_{1/2}$, having similar peak patterns and separated by an energy difference of ∼17 eV due to spin–orbit coupling.[67,68] The deconvolution of the Ni $2p_{3/2}$ region is complex with contributions from different species (oxide, hydroxide, and oxyhydroxide), nonlocal screening, and satellite features. The main peak at ∼854 eV corresponds to Ni in its +2 oxidation state, and the charge transfer $c3d^9L$ state (where c stands for a hole in the Ni 2p core level and L is a hole in the O ligand orbital). The broad features centered at

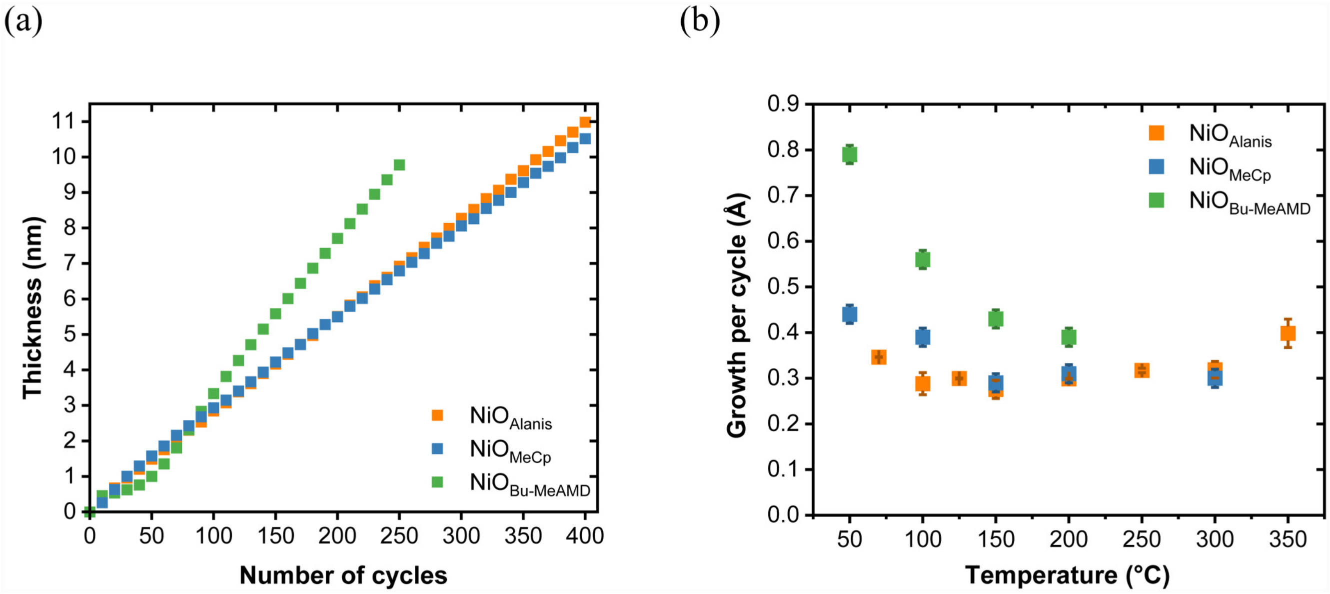

FIG. 3. (a) Film thicknesses of NiO_{Alanis}, NiO_{MeCp}, and $NiO_{Bu\text{-}MeAMD}$ as a function of number of ALD cycles deposited at 150 °C. (b) GPC of NiO_{Alanis}, NiO_{MeCp}, and $NiO_{Bu\text{-}MeAMD}$ at different deposition temperatures.

~861 and ~866 eV are satellite features.[2,6,68] The shoulder peak at ~856 eV has contributions from Ni^{2+} states from $Ni(OH)_2$ species, Ni^{3+} states, nonlocal screening effect, and surface cluster contributions.[68–70] Thus, this makes it challenging to determine the ratio of the Ni^{3+}-to-Ni^{2+} states from the Ni 2p spectra. The O1s spectrum of the NiO films, as shown in Fig. S6(b) in the supplementary material, indicates the presence of the Ni–O–Ni bond at ~529 eV and Ni–OH at ~531 eV. The latter can be attributed to the presence of nickel hydroxide [$Ni(OH)_2$] and nickel oxyhydroxide (NiOOH). Chemical species, such as NiOOH, further show the presence of Ni in its +3 oxidation state. The presence of hydroxyl groups is also evidenced by the hydrogen detected from the ERD measurements, which is found in metal oxides in the form of hydroxyl groups.[71] The presence of two peaks in Figs. S5(b) and S5(d) in the supplementary material suggests that different concentrations of hydrogen are present at the bulk and surface of the NiO_{Alanis} and NiO_{MeCp} films. The H content in the $NiO_{Bu\text{-}MeAMD}$ films is higher compared to the PA-ALD-based NiO films, indicating a higher hydroxyl content. This is further corroborated by the increase in the hydroxyl feature of the former observed in Fig. S6(b) in the supplementary material. Hydrogen can act as a donor center to the excess oxygen, thereby compensating for the Ni^{3+} acceptor states.[14,71]

All the NiO films exhibit low levels of carbon and nitrogen impurities, below the detection limit of RBS (100 atoms/nm^2). So, the impurity levels, reported in at. %, are measured by XPS after gentle sputtering of the NiO surface to remove the adventitious carbon layer. The $NiO_{Bu\text{-}MeAMD}$ films have a relatively higher impurity content compared to the PA-ALD NiO films due to the presence of residual ligands (see Table III). This incorporation of residual ligands in the $NiO_{Bu\text{-}MeAMD}$ film leads to an apparently thicker film, therefore resulting in higher GPC values. This is further corroborated by the lower mass density of the $NiO_{Bu\text{-}MeAMD}$ films compared to the NiO_{MeCp} and NiO_{Alanis} films, as reported in Table III. The higher mass density and the lower impurity content of the NiO_{MeCp} and NiO_{Alanis} films can be attributed to the film densification process occurring when ions impinge on the surface of the film during the O_2 plasma half-cycle.[73,74]

TABLE II. Comparison of the different ALD processes based on GPC and nonuniformity. The nonuniformity is calculated using Eq. (1), based on the thickness of the NiO films (see Fig. S4 in the supplementary material) deposited on a 4″ c-Si wafer and 8″ c-Si wafer for $NiO_{Bu\text{-}MeAMD}$.

	GPC at 150 °C (Å/cycle)	Nonuniformity (%)
NiO_{Alanis}	0.27 ± 0.02	6.7
NiO_{MeCp}	0.29 ± 0.01	1.1
$NiO_{Bu\text{-}MeAMD}$	0.43 ± 0.01	0.3
$NiO_{Bu\text{-}MeAMD}$ (8″ area)	0.43 ± 0.01	2.5

C. Structural and crystallographic properties

When comparing the morphology of the various NiO films, the top-view high-angle annular dark field (HAADF)-STEM images in Figs. 5(a)–5(c) show the lateral grain size of each film to be in the range of 5–7 nm. The grain boundaries are indicated by the darker contours surrounding the grains. In the $NiO_{Bu\text{-}MeAMD}$

TABLE III. Elemental composition of NiO_{Alanis}, NiO_{MeCp}, and $NiO_{Bu\text{-}MeAMD}$, deposited at 150 °C, as determined by RBS, ERD (for hydrogen), and XPS (for C and N). The C and N at. % are extracted from XPS. The mass density of NiO is calculated from RBS measurements and the thickness ("*t*") derived from the optical modeling of the films by spectroscopic ellipsometry and compared to that from XRR.

	C	N	Ni	O	H	Ni:O	t	Density	
	(at. %)		(atoms/nm^2_{geo})				(nm)	(g/cm^3)	
	from XPS							RBS	XRR
NiO_{Alanis}	∼1	∼1	1154 ± 23	1412 ± 71	134 ± 13	0.82 ± 0.04	22 ± 1	6.8 ± 0.8	6.8 ± 0.3
NiO_{MeCp}	∼1	—	1288 ± 26	1339 ± 67	184 ± 18	0.96 ± 0.05	23 ± 1	7.0 ± 0.9	7.1 ± 0.3
$NiO_{Bu\text{-}MeAMD}$	5 ± 1	∼1	1009 ± 20	1083 ± 54	263 ± 18	0.93 ± 0.05	25 ± 1	5.1 ± 0.5	5.4 ± 0.5

film, areas of lower thickness or porosity can be recognized, indicated by the presence of darker areas as shown in Fig. 5(c). However, it cannot be concluded whether these correspond to pinholes extending to the substrate. These areas are absent in NiO_{Alanis} [Fig. 5(a)] and NiO_{MeCp} [Fig. 5(b)]. The bright field (BF) cross-sectional TEM images of the NiO films, presented in Figs. 5(d)–5(f), indicate that the layers are crystalline. Notably, columnar grains are present in the case of NiO_{MeCp} and NiO_{Alanis} films, stretching over the entire thickness. In contrast, in $NiO_{Bu\text{-}MeAMD}$ films, differences in diffraction contrast reveal that some of the grains are not continuous over the entire thickness, as shown in Fig. 5(f) and Fig. S9 in the supplementary material. Moreover, the presence of smaller grain size and partial overlap of these grains within the lamella thickness leads to higher contrast variation within the $NiO_{Bu\text{-}MeAMD}$ layer compared to the other films in the HAADF-STEM cross-sectional TEM images [see Fig. S10(c) in the supplementary material]. This makes it challenging to distinguish the contribution to contrast variation from any potential porosity present in the film from the HAADF-STEM cross-sectional images.

The crystallinity of the NiO films, as determined by GI-XRD, is shown in Fig. S11 in the supplementary material. The NiO thin films display a rock salt (face-centered cubic) crystallographic phase. All diffraction patterns show the set of (111), (200), (220), (311), and (222) reflections that are expected for a poly-crystalline film although the peak ratios differ among the NiO films as well as from the powder diffraction pattern displayed as a reference. Furthermore, Bragg–Brentano (θ–2θ) scans are carried out to gain more insight into the preferential orientation of the grains since, in this geometry, only crystallographic planes oriented parallel to the sample surface contribute

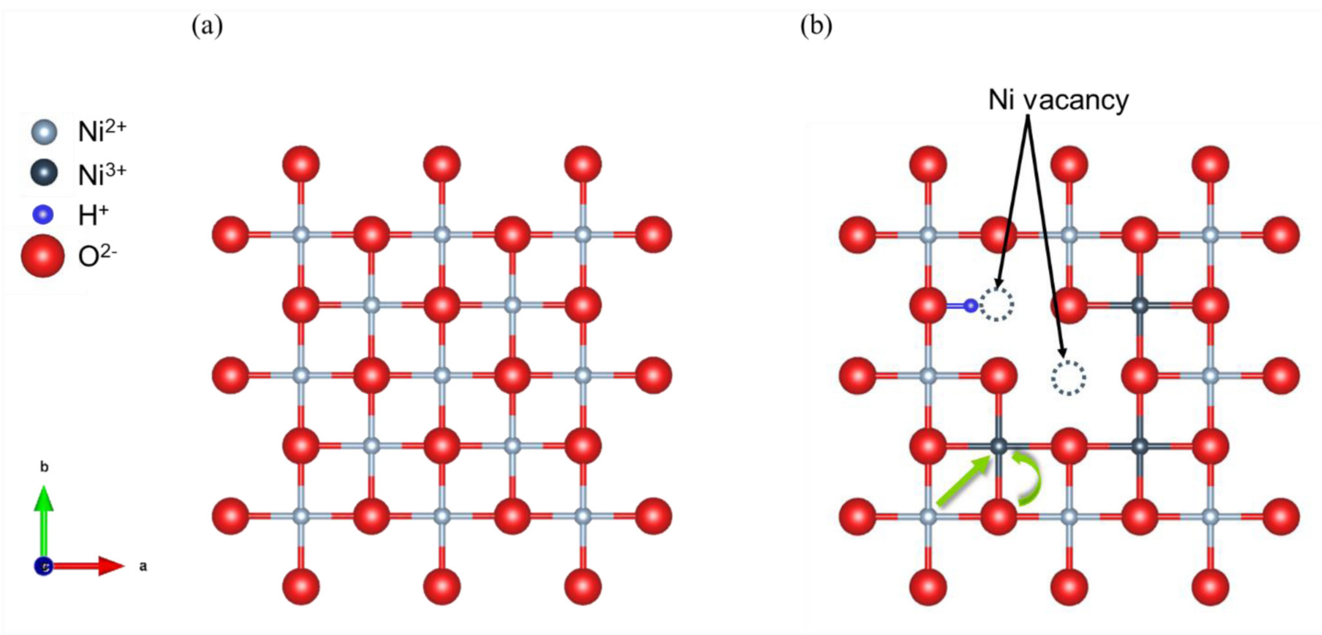

FIG. 4. Schematic of the crystal lattice of (a) stoichiometric NiO and (b) nonstoichiometric NiO containing Ni vacancies, Ni^{3+} states, and –OH groups. The arrow indicates electron transfer from Ni^{2+} or O^{2-} to Ni^{3+} acceptor states. The crystal lattice is visualized using the VESTA software (Ref. 72).

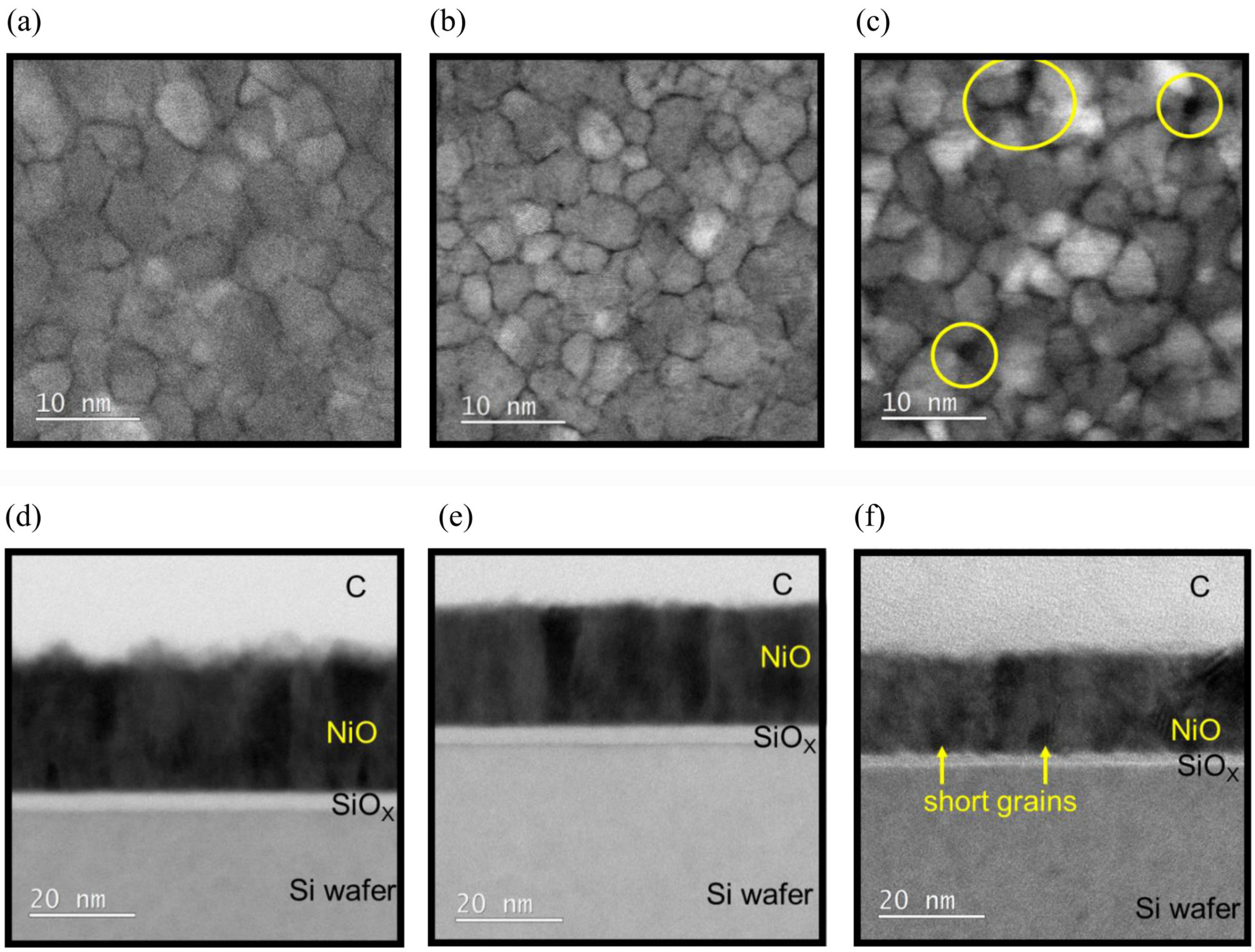

FIG. 5. Top-view (top) HAADF scanning TEM images of (a) NiO_{Alanis} (~11 nm), (b) NiO_{MeCp} (~9 nm), and (c) $NiO_{Bu\text{-}MeAMD}$ films (~8 nm). Some of the areas with a lower film thickness in the $NiO_{Bu\text{-}MeAMD}$ films are marked by circles. Cross-sectional BF STEM images of (d) NiO_{Alanis}, (e) NiO_{MeCp}, and (f) $NiO_{Bu\text{-}MeAMD}$ films, respectively.

to the diffractogram, providing information on the out-of-plane texture of the films.[75,76] As shown in Fig. 6, the XRD diffractograms suggest a ⟨100⟩ texture for the NiO_{Alanis} and NiO_{MeCp} films and a ⟨111⟩ texture for $NiO_{Bu\text{-}MeAMD}$ films. The preferential growth of $NiO_{Bu\text{-}MeAMD}$ along the ⟨111⟩ direction can be initiated by the presence of a higher content of hydroxyl groups in the $NiO_{Bu\text{-}MeAMD}$ film, which has been observed to stabilize the NiO (111) surface.[77,78] The lattice constants of the NiO_{Alanis}, NiO_{MeCp}, and $NiO_{Bu\text{-}MeAMD}$ films are 4.17 ± 0.01, 4.18 ± 0.01, and 4.19 ± 0.01 nm, respectively, and agree with previous reports on NiO films.[39,79]

From an application perspective, the crystal orientation of NiO films has been reported to play a crucial role in determining their electrochemical performance, particularly in the oxygen evolution reaction (OER).[24,80] The NiO (111) surface is polar, terminating with either Ni or O atoms, whereas the NiO (200) surface is nonpolar, terminating with both Ni and O atoms. This difference in surface termination influences the surface energy of the films.[3,80] Khamene *et al.* demonstrated that ALD NiO films with ⟨111⟩ texture are a more OER-active catalyst than NiO with ⟨200⟩ texture, owing to a higher hydroxide coverage and higher surface energy. The higher surface energy correlates with increased reactivity and promotes stronger interaction between the catalyst and reaction intermediates.[24] Additionally, the presence of the porous surface structure of the $NiO_{Bu\text{-}MeAMD}$ films can increase the accessibility of active sites for OER.[7]

D. Optoelectrical properties

One of the most important material properties of NiO relevant to its many applications, particularly in solar cells, as a hole

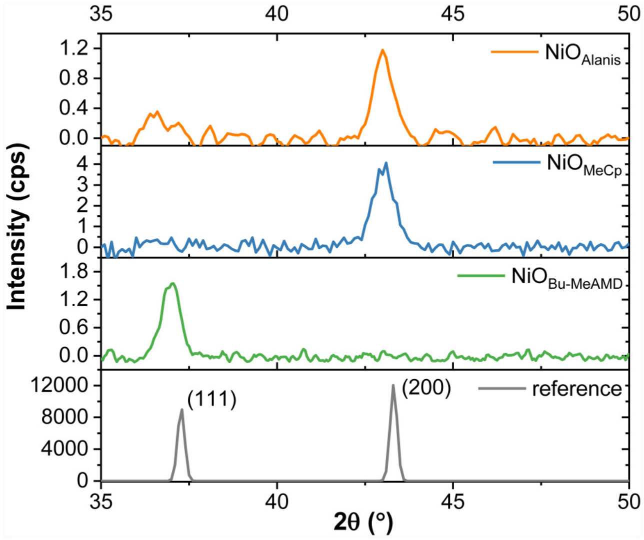

FIG. 6. Bragg–Brentano (θ–2θ) scans diffractogram of the NiO films with a thickness of ∼20 nm. The reference diffractogram of NiO powder is obtained from the ICSD database (collection code–9866).

transport layer[1,3] and in gas sensors,[9] is its conductivity. All NiO films exhibit p-type conductivity, as evidenced by the negative slope of the Mott–Schottky plots shown in Fig. S12 in the supplementary material.[81] The lateral resistivity of the NiO films, measured by a four-point probe, is reported in Table IV. The resistivity of the NiO_{Alanis} is lower than that of NiO_{MeCp} and $NiO_{Bu\text{-}MeAMD}$, being more Ni-deficient than the other NiO films.

Additionally, the charge carrier concentration of the NiO films is inferred from the Mott–Schottky analysis. As reported in Table IV, the NiO_{Alanis} films exhibit the highest carrier concentration among the NiO samples, consistent with their low Ni-to-O ratio, which likely contributes to the observed low resistivity. Furthermore, the Fermi level is expected to be closer to the valence band for a p-type material as the doping level increases. The presence of a higher carrier concentration in the NiO_{Alanis} film is further corroborated by UPS measurements, shown in Figs. S14 and S17 in the supplementary material, which reveal that the Fermi level lies closer to the valence band in the NiO_{Alanis} films as compared to other films.

Next to this, film morphology can affect the electrical properties, particularly mobility. The carrier mobility of the NiO films could not be measured by the Hall effect, suggesting a low hole mobility below the measurement limit (1 cm^2/Vs). Based on the carrier concentration, the carrier mobility of the NiO films is estimated to be in the range of $\sim 10^{-5}\,cm^2\,V\,s^{-1}$ ($NiO_{Bu\text{-}MeAMD}$) to $10^{-3}\,cm^2\,V\,s^{-1}$ (NiO_{Alanis}). A similar range of carrier mobility values of NiO films has been reported in the literature, varying with the synthesis route.[82,83] The lower mass density of the thermal ALD $NiO_{Bu\text{-}MeAMD}$ films indicates the presence of porosity within the films. This can negatively impact the charge carrier mobility.[84] Additionally, while the lateral grain size of the NiO films is comparable (as shown in Fig. 5), $NiO_{Bu\text{-}MeAMD}$ films, unlike NiO_{Alanis} and NiO_{MeCp} films, exhibit grains that do not continuously extend vertically, resulting in multiple grain boundaries. This can hinder charge transport by acting as scattering centers, thereby reducing carrier mobility.[85,86]

Looking back at the application of NiO films in organic solar cells[87] and perovskite solar cells,[88] studies have shown that lowering the resistivity can improve the device performance, particularly the fill factor (FF) and open-circuit voltage (V_{OC}). Decreasing the resistivity has been shown to lead to a lower series resistance of the solar cell and lower recombination due to reduced hole accumulation at the NiO/absorber interface, thereby increasing the FF and V_{OC}, respectively.[87,88] However, the gain in V_{OC} is modest and shadowed by the high interface recombination at the NiO-based devices due to the presence of Ni^{3+} states on the surface, which have been reported to act as trap states.[3] Moreover, Boyd *et al.* reported that the Ni^{3+} surface states can undergo a redox reaction with the organic cation of the perovskite absorber, forming a dead layer and detrimentally affecting the performance of NiO-based perovskite solar cells.[89] This highlights the complex interplay of the Ni^{3+} states on the resistivity and, when present at the surface, underscoring the need for tailoring NiO properties to increase device performance.

In parallel, the presence of Ni^{3+} states has been observed to affect the optical properties of the NiO films.[90,91] The transmittance of the NiO films on glass decreases in the visible region (400–800 nm) upon a decrease in NiO film resistivity, with the $NiO_{Bu\text{-}MeAMD}$ having the highest transmittance, as shown in Fig. 7(a). The lower transmittance and resistivity of the NiO_{Alanis} film compared to the NiO_{MeCp} films can be attributed to the precursor rather than the difference in the plasma exposure time during the O_2 plasma half-cycle. To assess the impact of the total plasma exposure, Alanis™-based NiO films are also deposited using 3 s O_2 plasma exposure, similar to the NiO_{MeCp} process. These films exhibit resistivity, transmittance (Fig. S18 in the supplementary material), film density, and relative impurity content, similar to the films deposited with 5 s O_2 plasma exposure

TABLE IV. Comparison of electronic and optical properties of the (∼9 nm) NiO films. The transmittance of NiO films deposited on glass is reported here in the wavelength range of 400–800 nm. The transmittance of the bare glass substrate is 91% in this wavelength range.

	Resistivity (Ω cm)	Carrier concentration (cm^{-3})	VBM.–E_F (eV)	Transmittance (%)	Bandgap (eV)
NiO_{Alanis}	6 ± 2	$(3.1 \pm 0.2) \times 10^{20}$	0.62 ± 0.05	79	3.66 ± 0.02
NiO_{MeCp}	66 ± 33	$(1.9 \pm 0.1) \times 10^{20}$	0.76 ± 0.05	84	3.69 ± 0.02
$NiO_{Bu\text{-}MeAMD}$	$(1.7 \pm 0.1) \times 10^4$	$(2.8 \pm 0.2) \times 10^{19}$	0.87 ± 0.05	90	3.82 ± 0.02

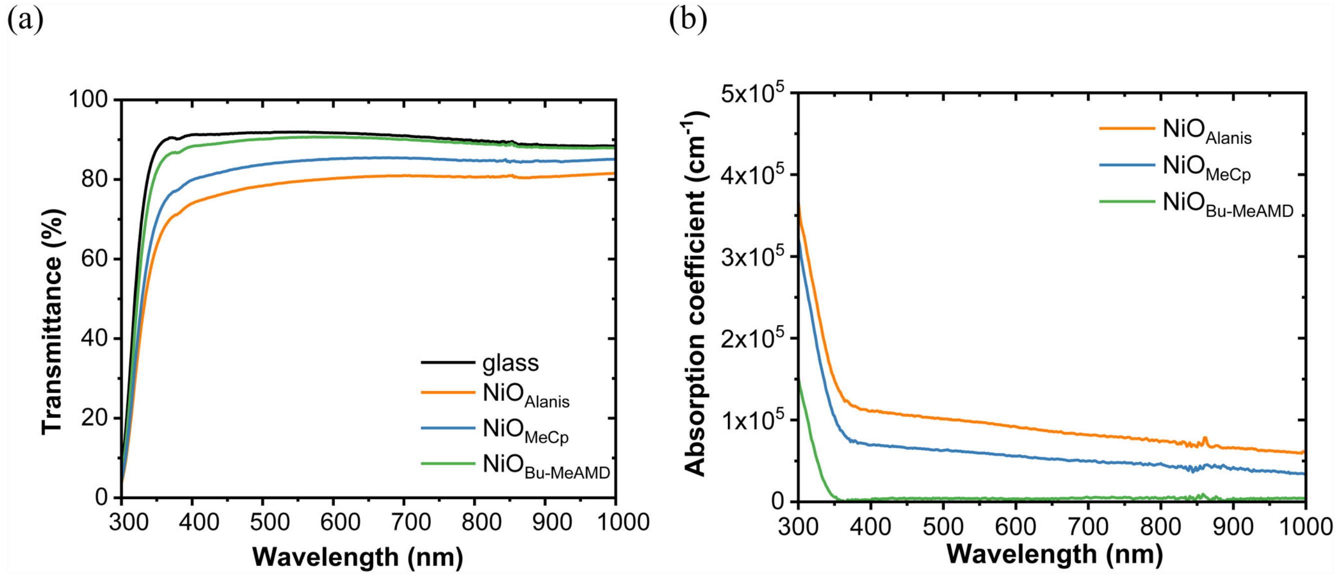

FIG. 7. (a) Transmittance and (b) absorption coefficient of ~9 nm NiO_{Alanis}, NiO_{MeCp}, and $NiO_{Bu\text{-}MeAMD}$.

(see Table SIII in the supplementary material). The decrease in transmittance of NiO_{MeCp} and NiO_{Alanis} films is mainly due to the absorption by these films in the visible spectrum because the reflection by all the NiO films is similar, as shown in Fig. S15 in the supplementary material. This is supported by the continuous background absorption in the order of nearly $\sim 10^5\,cm^{-1}$ of the NiO_{MeCp} and NiO_{Alanis}, as compared to $NiO_{Bu\text{-}MeAMD}$, in the visible spectrum [Fig. 7(b)]. Miedzinska *et al.* attributed the reduced transmittance of nonstoichiometric NiO to the presence of Ni^{3+} acceptor states and hydroxides, leading to the formation of defect states near the valence band edge.[90,92] While transitions from these defect states are typically forbidden, they occur due to phonon interaction, resulting in low-intensity sub-bandgap absorption. These defect states exhibit several transitions with energies ranging from 1.1 to 3.5 eV, leading to a continuous background absorption.[64,90–92] To conclude, a transmittance loss of ~11% in the wavelength range of 400–800 nm is observed for NiO_{Alanis} compared to $NiO_{Bu\text{-}MeAMD}$ on lowering the resistivity by a factor of $\sim 10^4$. When employed as a hole transport layer in p-i-n perovskite solar cells, a decrease in the transmittance of NiO can affect the short-circuit current density (J_{SC}). For instance, Zhang *et al.* observed that increasing the transmittance of the NiO film from 60% to 80% in the visible region increased the J_{SC} by $\sim 2\,mA/cm^2$.[93]

The optical bandgap ("E_g") of the NiO films is calculated from the Tauc plot using the absorption coefficient derived from UV-visible measurements, as shown in Fig. S16 in the supplementary material. The band structure of stoichiometric NiO, as determined by density functional theory calculations, reveals that it has an indirect bandgap,[94] but the presence of Ni^{3+} acceptor states near the valence band maxima results in a direct optical transition.[95] In this study, the direct bandgap of the NiO films is estimated from the steep absorption onset in the Tauc plots (see Fig. S16 in the supplementary material). Since the NiO films have sub-bandgap absorption, the bandgap is estimated from the intersection of the Tauc line with a baseline fitting the sub-bandgap absorption region.[96,97] As can be seen in Table IV, the optical bandgap decreases with a decrease in resistivity, with the $NiO_{Bu\text{-}MeAMD}$ having a wider bandgap than NiO_{MeCp} and NiO_{Alanis}. Wrobel *et al.* demonstrated that hole doping alters the electronic band structure of NiO, leading to the narrowing of the bandgap as the dopant concentration increases.[64,98]

Figure S17 in the supplementary material presents a schematic energy-level diagram of the NiO films. The valence band edge aligns well with many commonly adopted perovskite[99] and organic absorbers,[100] facilitating efficient hole extraction, while the conduction band is sufficiently high to block electron transfer. Nevertheless, band alignment also depends on the chemical composition of the absorber, and some absorbers, such as certain narrow bandgap perovskites with high valence band maximum, may not align well.[101] This indicates that all the ALD NiO films are ideally suitable as hole transport layers in organic and perovskite solar cells, as well as in tandem photovoltaics. However, as discussed earlier, the presence of Ni^{3+} states at the NiO surface may negatively impact the device performance.

IV. CONCLUSION

This study presents a comprehensive comparison of NiO films from three ALD processing routes, including a newly developed PA-ALD process of NiO using the novel Alanis™ precursor. The NiO_{Alanis} process exhibits a linear growth with a GPC of 0.27 ± 0.02 Å/cycle in the temperature window of 100–250 °C. The NiO_{MeCp} process has a similar GPC, whereas the $NiO_{Bu\text{-}MeAMD}$

process is characterized by a higher GPC of 0.43 ± 0.01 Å/cycle at 150 °C. The presence of a relatively larger impurity content due to the residual precursor ligands can lead to the observed higher GPC value. Furthermore, the presence of energetic ions from the O_2 plasma impinging on the NiO surface leads to a denser PA-ALD NiO film, as compared to the thermal $NiO_{Bu\text{-}MEAMD}$ films.

All the NiO films are substoichiometric in nature, with the NiO_{Alanis} film having the lowest Ni-to-O ratio, pointing to a higher relative content of Ni^{3+} acceptor states. Moreover, the higher H content in the $NiO_{Bu\text{-}MeAMD}$ film, compared to the PA-ALD NiO films, can also lower the Ni^{3+} content in the former film. This results in a lower resistivity of the NiO_{Alanis} film as compared to NiO_{MeCp} and $NiO_{Bu\text{-}MeAMD}$. Furthermore, the presence of porosity, as suggested by the lower density of the $NiO_{Bu\text{-}MeAMD}$ films and a relatively higher density of grain boundaries, can affect the carrier mobility, thereby leading to the high resistivity observed in the films. On the other hand, the presence of a relatively larger Ni^{3+} content in the NiO_{Alanis} leads to absorption in the visible region due to sub-bandgap absorption by the Ni^{3+} states and a lower transmittance as compared to NiO_{MeCp} and $NiO_{Bu\text{-}MeAMD}$. To conclude, the ALD process affects the morphology and chemical composition of the NiO films, thereby influencing the optical and electronic properties of the NiO films.

Also, reflecting on the properties of NiO films in the context of their applications, it is worth reporting that the presence of Ni^{3+} states not only affects the film's optoelectrical properties but also impacts the implementation of NiO films in devices. For example, Ni^{3+} states on the surface of NiO films, serving as hole transport layers, can lead to trap-assisted nonradiative recombination and redox reactions at the NiO/perovskite absorber interface. This suggests that NiO films with a lower Ni-to-O ratio, such as NiO_{Alanis}, may exhibit a lower performance when implemented as a hole transport layer, as compared to devices based on the other NiO films. Beyond Ni^{3+} states, the crystallographic orientation of NiO films can also influence their performance in specific applications, such as electrocatalysis. Notably, NiO films exhibiting a $\langle 111 \rangle$ texture, such as $NiO_{Bu\text{-}MeAMD}$ films, demonstrate a higher OER activity compared to NiO films with a $\langle 200 \rangle$ texture due to their higher hydroxide coverage and higher surface energy. These findings underscore the need for the tailoring of NiO material properties, with careful consideration of their implications, to optimize performance in specific applications.

SUPPLEMENTARY MATERIAL

See the supplementary material for specific details about ALD processing, thickness uniformity mapping, XPS, XRR, UPS, reflection data, Mott–Schottky plot, and Tauc plot of the NiO films.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Kousumi Mukherjee: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Cristian van Helvoirt:** Data curation (equal); Formal analysis (equal); Methodology (lead); Validation (equal); Writing – review & editing (equal). **Christ H. L. Weijtens:** Data curation (equal); Formal analysis (equal); Resources (equal); Writing – review & editing (equal). **Marcel Verheijen:** Data curation (equal); Formal analysis (equal); Resources (equal); Writing – review & editing (equal). **Mariadriana Creatore:** Conceptualization (equal); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support this article have been included as part of the ESI.

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